Question 4 JOINS, Filtering, AGGREGATES

Table 1; Employees

Employee ID	First Name	Last Name	Department	Salary	City Age	Age
1	John	Doe	IT	70000	Newyork	35
2	Alice	Smith	HR	60000	Los Angeles	40
3	Bob	Johnson	Finance	75000	Toronto	45
4	David	Brown	IT	72000	London	30
5	Emma	Wilson	Sales	65000	Sydney	28
6	Michael	Clark	Finance	80000	New York	50

Table 2: Projects

ProjectID	EmployeeID	ProjectName	Budget	StartDate	Status
101	1	AI development	100000	2024-01-10	Completed
102	2	Employee Training	50000	2024-02-5	Ongoing
103	1	CyberSecurity Audit	75000	2024-03-05	Pending
104	3	Financial Analysis	90000	2024-04-12	Ongoing
105	5	Market Expansion	65000	2024-05-20	Completed
106	6	Risk Management	8000	2024-06-10	Pending

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Questions 4

1. SQL Joins

Retrieve all employees and their assigned projects using an INNER JOIN

SQL

```
SELECT employee-id, firstname, Last-name, Department, Salary
        city, Age
FROM Employee AS A
INNER JOIN projects AS B
ON A.employee-id = B.employee-id;
```

Employee ID	Firstname	Lastname	Department	Salary	Project ID	Project NAME
1	John	Doe	IT	70000	101	AI Development
~~2~~ 2	Alice	Smith	HR	60000	102	Employee Training
~~3~~ 1	John	Doe	IT	70000	103	Cybersecurity Audit
3	Bob	Johnson	Finance	75000	104	Financial Analysis
5	Emma	Wilson	Sales	65000	105	Market Expansion
6	Micheal	Clark	Finance	80000	106	Risk Management

2. Retrieve all employees and their assigned projects, including employees who have no projects using a LEFT JOIN

Question 2 - LEFT JOIN

SQL

```
SELECT employeeID, FirstName, LastName, Department, Salary
        projectID, ProjectName, Budget, Status
FROM Employees AS A
 LEFT JOIN Projects AS B
  ON A. EmployeeID = B. EmployeeID;
```

outcome Table

EmployeeID	FirstName	LastNAME	Department	Salary	ProjectID	Project NAME
1	John	Doe	IT	70000	101	AI development
1	John	Doe	IT	70000	103	Cybersecurity Audit
2	Alice	Smith	HR	60000	102	Employee Training
3	Bob	Johnson	Finance	75000	104	Financial Analysis
4	David	Brown	IT	72000	Null	Null
5	Emma	WILSON	Sales	65000	105	Market Expansion
6	Micheal	Clark	Finance	80000	106	Risk Management

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3. Retrieve all projects and their assigned employees including project that have no employees using RIGHT JOIN

Question 3

SQL

SELECT ProjectID, ProjectName, Budget, Status, EmployeeID, FirstName, LastName, Department, Salary FROM Employees AS A RIGHT JOIN Project AS B

ProjectID	Project Name	Budget	Status	Employee ID	First Name	Last Name
101	AI Development	100000	Completed	1	John	Doe
102	Employee-Training	50000	Ongoing	2	Alice	Smith
103	Cyber Security Audit	75000	Pending	1	John	Doe
104	Financial Analysis	90000	Ongoing	3	Bob	Johnson
105	Market Expansion	65000	Completed	5	Emma	Wilson
106	Risk Management	80000	Pending	6	Michael	Clark

SQL

SELECT ProjectID, ProjectName, Budget, Status, EmployeeID, FirstName Lastname, Department, Salary

FROM Employees AS A
RIGHT JOIN Projects AS B
ON A.employeeID = B.EmployeeID;

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Case 4. FULL OUTER JOIN

Retrieve all employees and projects, including those without a match in either table using a Full outer Join

SQL

```
SELECT employeeID, FirstName, LastName, Department, Salary,
       Project_id, projectName, Budget, Status
FROM Employees AS A
FULL OUTER JOIN Projects AS B
  ON A.employeeID = B.Employee ID;
```

EmployeeID	FirstName	LastName	Department	Salary	Project_ID	Project Name	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	CyberSecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	Null	Null	Null	Null
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Micheal	Clark	Finance	80000	106	Risk Management	80000	Pending

3. Filtering Statements

Question 7 - Employees earning more than 70 000

SQL

```
SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
WHERE Salary > 70 000;
```

Outcome Table

Employee ID	First Name	Last Name	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Micheal	Clark	Finance	80 000

Question 8 - Employees in IT and Finance

SQL

```
SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
WHERE Department IN ('IT', 'Finance');
```

Outcome Table

Employee ID	First Name	Last Name	Department	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Micheal	Clark	Finance	80000

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Question 9 - Projects not Completed

SQL

```
SELECT ProjectID, ProjectName, Budget, Status
FROM Projects
WHERE Status <> 'Completed';
```

Outcome Table

ProjectID	ProjectName	Budget	Status
102	Employee Training	50000	Ongoing
103	CyberSecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

Question 10 - Budget > 70000 and not completed

SQL

```
SELECT ProjectID, ProjectName, Budget, Status
FROM Budget > 70000 AND Status <> 'Completed';
```

Outcome Table

ProjectID	ProjectName	Budget	Status
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

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Question 11 - Employees from New York or Toronto ordered by salary DESC

SQL

```
SELECT EmployeeID, FirstName, LastName, Department, Salary,
       City
FROM employees
WHERE city IN ('New York', 'Toronto')
ORDER BY Salary DESC;
```

~~Outcom~~ outcome table

Employee ID	FirstName	LastName	Department	Salary	City
6	Micheal	Clark	Finance	80000	New York
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	70000	New York

Question 12 - Top 3 Highest Paid employees

SQL

```
SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
ORDER BY Salary DESC
LIMIT 3;
```

Employee ID	FirstName	LastName	Department	Salary
6	Micheal	Clark	Finance	80 000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000

4. Aggregate Functions with GROUP BY AND HAVING

Question 13 - Total salary per department

SQL

```
SELECT Department,
       SUM(salary) AS TotalSalary
FROM Employees
GROUP BY department
ORDER BY TotalSalary DESC;
```

Department	TotalSalary
Finance	155000
IT	142000
Sales	65000
HR	60000

Question 14 - Average Salary per city greater than 65000

SQL

```
SELECT city
       AVG(Salary) AS AverageSalary
FROM Employees
GROUP BY city
HAVING AVG(Salary) > 65000;
```

City	Average Salary
New York	75000
Toronto	75000
London	72000

Question 15 - Departments with More than 1 employee

SQL

```
SELECT Department
       COUNT (EmployeeID) AS Employee Count
FROM Employee
GROUP BY Department
HAVING COUNT (Employee ID) > 1;
```

Department	Employee Count
IT	2
Finance	2

Question 16 - Project Statuses with at least 2 projects

SQL

```
SELECT Status
       COUNT (Project ID) AS Project Count
FROM Projects
GROUP BY status
HAVING COUNT (Project ID) >=2;
```

Status	Project Count
Completed	2
Ongoing	2
Pending	2

Question 17 - Employee with total project budget more than 150,000

SQL

```
SELECT Employee ID,
       First Name,
       Last Name
   SUM ( Budget) AS Total ~~Projecting~~ Project Budget
FROM Employees AS A
 INNER JOIN Projects AS B
 ON A. Employee ID = B. Employee ID
 GROUP BY Employee ID, First Name, Last Name
 HAVING SUM ( Budget ) > 150 000 ;
```

Employee ID	First Name	Last Name	Total Project budget
1	John	Doe	175000